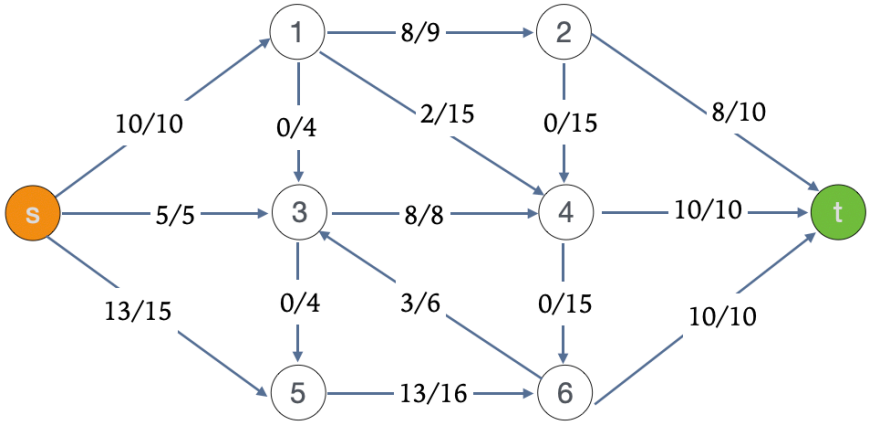
CPE 231 – Algorithms (1/69)

Homework III

1. (10 points) From the following maximum flow, derive the residual graph and then determine the minimum cut as well as the the corresponding cut capacity.

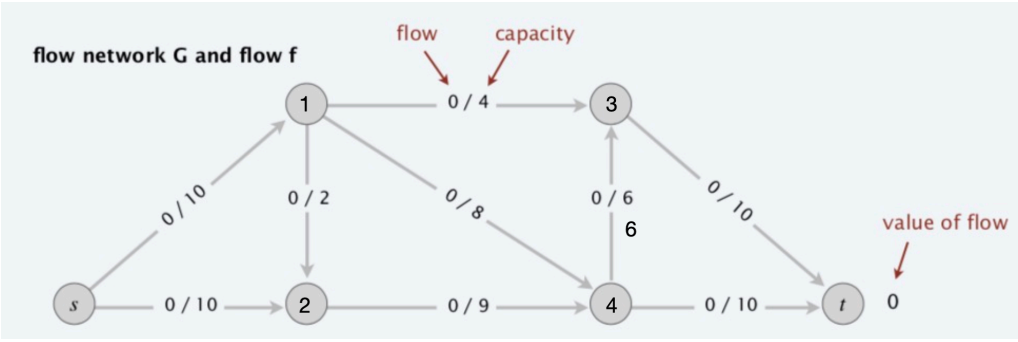


2. (20 points) Write a C program that accepts an input file containing the number of nodes N in the first line and a list of triplets (i, j, c_{ij}) where (i, j) is a directed edge and c_{ij} is the edge capacity in the remaining lines. The node labels must be between 0 to $N - 1$, where node 0 is the source and node $N - 1$ is the terminal (sink).

   Use the input to construct an adjacency list with N head entries to represent the network flow graph. Each element in the adjacency list represents an edge in the flow / residual network.

   From the adjacency list, implement the Ford-fulkerson method to find the maximum st-flow from source to sink with the augmenting paths being determined from breadth first search. Then, print out the max flow value and the flow assigned to each edge.

   Submit the source code and the output of your program for the flow network in the figure below.



3. (10 points) From the flow network in the Bike sharing in-class assignment 6-2,

   (a) (5 points) Use your program in Question 1 to solve the max-flow problem to determine the number of bikes to be transported among locations. Show the program output and draw the flow network solution.

   (b) (5 points) Are the customer reservations satisfied at every location ? Explain.